Midterm Project

Analyzing Technology Usage and Lifestyle Habits on Stress Levels

1. Dataset Creation

For this midterm project, We created a dataset focusing on analyzing the impact of technology usage and lifestyle habits on stress levels. We explored online sources, including Kaggle examples, to design a realistic and structured dataset. The dataset consisted of 1000+ rows records with four feature columns (at least one categorical) and one target column (categorical), following dataset requirements properly.

The features included:

- **Screen Time** (hours per day) — numerical
- **Bedtime** (hour of sleep) — numerical (24-hour format)
- **Age** (in years) — numerical
- **App Type Most Used** (categorical: social, entertainment, educational, gaming, utility)

The target column was:

- **Stress Level** (categorical: low, moderate, high)

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2. Data Preprocessing

The preprocessing steps performed are detailed below:

2.1 Handling Missing Values

Initially, missing values were manually introduced into the dataset, totaling 250 missing values in 5 columns.. These missing values were first marked as `NA`.
For handling missing values:

- **Numerical columns** (Screen Time, Bedtime, Age) were filled using **median imputation**, where missing values were replaced.
- **Categorical column** (App Type Most Used) was filled using **mode imputation**, replacing missing values with the most frequently occurring category.

- **Rows with missing values in the target column** (Stress Level) were **removed** to ensure the dataset only contains complete records for the target variable.

After imputation, there were 0 missing values left in the dataset.

2.2 Data Type Conversion

Next, data type conversion was performed to prepare the dataset for further analysis and modeling.

- The **App Type Most Used** column and the **Stress Level** column (target) were encoded using **Label Encoding**.
- Categories were assigned numerical labels (e.g., 1, 2, 3, 4 for app types, and numbers for stress levels).

This ensured that the categorical features were properly transformed into numerical format suitable for modeling.

2.3 Outlier Detection and Handling

Outliers were detected using the **IQR method**:

- **Screen Time**: 13 outliers were detected (users with very high compared to the average).
- **Bedtime**: 0 outliers detected.
- **Age**: 0 outliers detected.

To handle outliers, **capping** was applied instead of removing records.
This method adjusted the extreme outlier values by replacing them with upper threshold limits, maintaining all records while controlling the influence of outliers.

2.4 Data Transformation

Finally, **scaling** was applied to the numerical columns (`Screen Time`, `Bedtime`, `Age`).
Standard scaling (mean = 0, standard deviation = 1) was used to normalize the data, ensuring that all numerical features were on the same scale and contributed equally to future analysis.

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3. Key Findings from analysis

- Missing values were successfully handled using what's suitable approach for each column.
- Categorical data was properly converted into numerical form using label encoding.
- Outliers were detected mainly in screen time and were corrected by applying capping.
- Numerical features were scaled, making the data consistent and ready for further analysis or modeling.

4. Justification for Outlier Handling and Data Transformation

Outlier handling was necessary because extreme values, especially in screen time, could have heavily skewed the results of the analysis. Instead of removing outlier records, We used capping to limit their effect while keeping all observations in the dataset. This approach preserved the natural variation in user behavior without letting extreme values dominate the patterns.

Data transformation through scaling was essential because the numerical features — screen time, bedtime, and age — were measured on different scales. Without scaling, features with larger values could have influenced the results unfairly. Standard scaling brought all numerical features to a common scale, ensuring that each feature contributed equally during any future modeling or analysis.

Thank you